Output:

2. Implement vaccum-cleaner Agent using reflex and goal based agent:

```
--- Running REFLEX agent ---
Start state: location=B, status={'A': 'Dirty', 'B': 'Dirty'}
Step 1: percept=('B', 'Dirty') -> action=Suck
Step 2: percept=('B', 'Clean') -> action=Left
Step 3: percept=('A', 'Dirty') -> action=Suck
Goal reached! Final status: {'A': 'Clean', 'B': 'Clean'}

--- Running GOAL agent ---
Start state: location=B, status={'A': 'Clean', 'B': 'Dirty'}
Step 1: percept=('B', 'Dirty') -> action=Suck
Goal reached! Final status: {'A': 'Clean', 'B': 'Clean'}
PS D:\AI>
```

1. Implement TIC-TAC-TOE game:

```
Tic Tac Toe - you are X, the computer is O.

 1 | 2 | 3
---+---+---
 4 | 5 | 6
---+---+---
 7 | 8 | 9

Your move (1-9): 1

 X | 2 | 3
---+---+---
 4 | 5 | 6
---+---+---
 7 | 8 | 9

Computer is thinking...

 X | 2 | 3
---+---+---
 4 | O | 6
---+---+---
 7 | 8 | 9

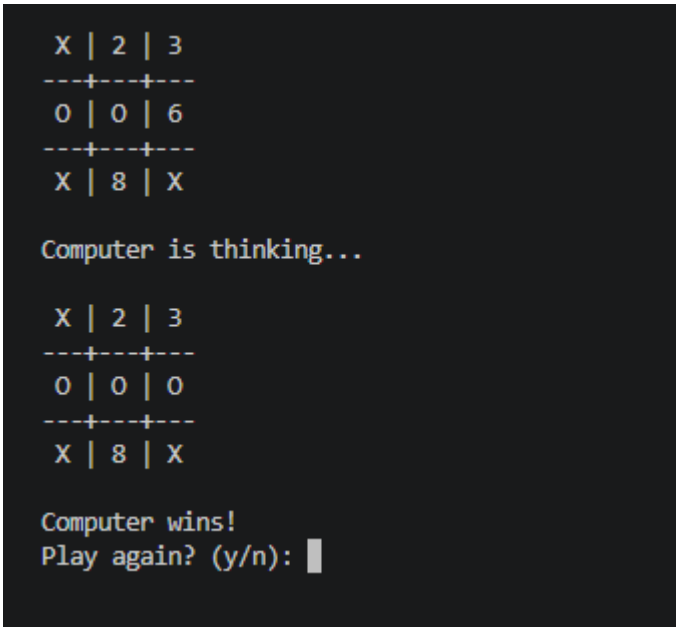
Your move (1-9): 7

 X | 2 | 3
---+---+---
 4 | O | 6
---+---+---
 X | 8 | 9

Computer is thinking...

 X | 2 | 3
---+---+---
 O | O | 6
---+---+---
 X | 8 | 9

Your move (1-9): 1
That square is already taken.
Your move (1-9): a
Please enter a number from 1 to 9.
Your move (1-9): 15
Please enter a number from 1 to 9.
Your move (1-9): 9
```


X | 2 | 3
---+---+---
O | O | 6
---+---+---
X | 8 | X
Computer is thinking...
X | 2 | 3
---+---+---
O | O | O
---+---+---
X | 8 | X
Computer wins!
Play again? (y/n):